ETHAN C. GOLDSTEIN

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Computer Information Systems student who designs and operates autonomous agent infrastructure. Built Autonomous OS, an 11-agent orchestration platform of roughly 12,700 lines of Node, TypeScript, and React that runs unattended on self-hosted hardware against live third-party APIs, with prompt-injection guardrails on every model call and a human approval gate before every outbound action.

EDUCATION

University of South Carolina — *Columbia, SC*
B.S. Computer Information Systems, Minor in Business Information Management — *Expected May 2027*
Coursework: Data Structures and Algorithms (Java), Information Security Principles, Web Applications, Capstone Project
Study Abroad: Florence University of the Arts, Firenze, Italy — *Spring 2026*

EXPERIENCE

AI & Workflow Automation Engineer — *2025 to Present*
Independent Practice · github.com/ethan-goldstein/Autonomous-OS — *Remote*

- Designed an 11-agent orchestration platform on a Node and Express ESM core where every agent inherits scheduling, SQLite persistence, and live log streaming from a shared base class, so adding an agent is one file and one registry line.
- Built a two-tier LLM dispatcher, headless Claude Code CLI primary with local Ollama fallback behind a 10-minute persisted circuit breaker, and capped inference at 2 concurrent processes on a FIFO semaphore after unbounded forking crashed an 8GB machine.
- Persisted the fleet on node:sqlite with no ORM across 14 tables and 10 indexes, adding WAL journaling and a 5-second busy timeout after concurrent writes began failing boots, and streamed live state to a React and TypeScript dashboard over Server-Sent Events.
- Injected prompt-injection guardrails at a single chokepoint into every model call, treating fetched content as data and never instructions, and routed all egress through one approval queue where risky lanes require a human tap no toggle can override.
- Wrote HMAC-SHA256 session auth on node:crypto alone behind a prefix allowlist mounted before any route is declared, with no localhost bypass because Tailscale terminates TLS on loopback; deployed via launchd with no public port.
- Made automated outreach defensible: per-lane daily caps dripped across staggered windows, randomized send jitter, DNS MX pre-validation that fails open, and an IMAP bounce loop retiring dead addresses.
- Added a self-distilling memory layer where each run writes durable facts to a deduplicated table the whole fleet reads, plus a sweep re-parsing every cron expression to replay runs a sleep missed.

GovCIO — *2026 to Present*
Data Processor — *Tysons Corner, VA*

- Hold an active Public Trust clearance (U.S. Treasury, IRS) extended to Department of Veterans Affairs systems by background-investigation reciprocity, PIV-sponsored through GSA USAccess.
- Process confidential Veteran records under the VA HTMS contract in secured remote environments.

Oxford Government Consulting — *2025*
Data Entry Operator — *Tysons Corner, VA*

- Supported the IRS Digitalization-as-a-Service modernization program in partnership with GovCIO, on a digitization pipeline processing 2 million document images per day.

The Baseball Zone, *Manager*. Rebuilt the company website; ran front desk operations, scheduling, and daily finances. — *2021 to 2024*

PROJECTS

Real-Time Multiplayer Engine *WebSockets, Cloudflare Durable Objects, Three.js, MediaPipe*

- Authoritative game server on edge compute: fixed 60Hz simulation with seeded RNG decoupled from render, WebSocket sessions with reconnect, AI backfill for empty rooms. Three public titles.

Vector Stroke-Rendering Engine *React, Vite, Three.js, GitHub Actions*

- Auto-traces a raster logo into vector contours with marching squares and replays the path against a Three.js pen, code-split so it never enters the main bundle. Live on a paying client's site.

Degrading Media Pipeline *Next.js, React, TypeScript, GSAP*

- Asset manifest resolving each of 18 scenes to video, still image, or CSS gradient depending on what exists, so the build ships before any final asset lands.

SKILLS

Languages: Java, Python, TypeScript, JavaScript (ES modules), SQL, Bash, HTML, CSS
AI & Agents: multi-agent orchestration, local LLM inference (Ollama), Claude Code CLI, prompt-injection defense, human-in-the-loop approval gates, circuit breakers, concurrency control
Backend & Distributed Systems: Node.js, Express, WebSockets, Server-Sent Events, node:sqlite (WAL), REST, OAuth 2.0, HMAC-SHA256 auth, Cloudflare Workers and Durable Objects, Tailscale, launchd and cron, Playwright, Git, GitHub Actions CI/CD
Frontend: React, Next.js App Router, Vite, Three.js, WebGL, GSAP, Framer Motion, MediaPipe, code splitting, WCAG and reduced-motion accessibility
Clearance: Public Trust (Active), U.S. Department of the Treasury (IRS); VA reciprocity granted 2026